

© Kosan Crisplant 2003

General description

The CUC controlled electronic leak detector type ET1-PT, is designed for automatic check of the tightness of LPG cylinders with centre valves.

The detection system used in the ET1-PT is based on the pressure rise principle.

For centre valves in general it is possible to check the tightness at the seat. For centre valves type Compact it is possible to check the tightness at the seat and the tightness between the inner gasket and the regulator.

Features

- Fully automatic operation
- Automatic sorting out of leaky cylinders
- Minimising of the possibility of human errors
- High degree of safety for end user
- Minimum of maintenance
- Low operation costs
- Handling of cylinder batches with varying cylinder dimensions

Functional description

The cylinders are carried on the chain conveyor to a position under the leak detector, where a test head is lowered onto the valve.

A leak, if any, will cause a pressure rise. This will actuate a pressure transmitter which will send a signal to the CUC for automatic sorting out of the leaky cylinder. At the same time a red control lamp is lit.

However, if no leak is detected, the cylinder will continue on the chain conveyor.

Scope of supply

The leak detector is supplied complete, ready for installation in chain conveyor plants with a standard level of 550 mm.

Additional equipment

(Only included if specified in quotation)

- Additional test bells/heads for different valve/cylinder types
- Spare parts set

© Kosan Crisplant 2003

Technical data

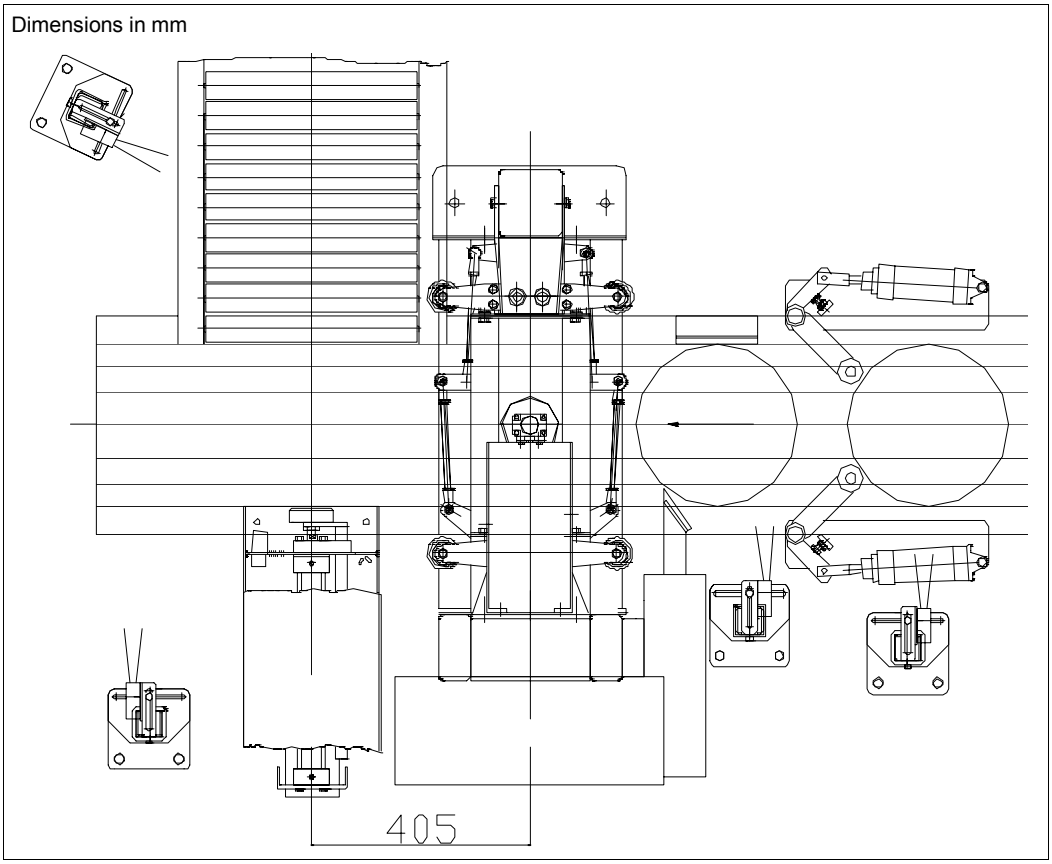
Capacity and conditions	• Max. 1200 cylinders/hour, depending on the following factors: – The chain conveyor speed – The number of sort outs – The cylinder type – The mixture of the gas (butane, propane or a mixture) – The leak rate
Standard sensitivity	• 2 grams of gas per hour
Cylinder dimensions (requirements and limitations)	• Cylinder diameter: – Min. 220 mm - Max. 400 mm – Max. diameter variation for all cylinders: 50 mm • Cylinder height: – Min. 200 mm - Max. 1000 mm – Max. height variation for all cylinders: 150 mm • Shroud diameter: – Inside diameter: Min. 100 mm – Outside diameter: Max. diameter of the cylinder
Necessary connections (specifications and quality requirements)	• Compressed air installation: – Connection size: G1/2" BSP – Working pressure: Min. 0.6 MPa. (6 bar) - Max. 1.0 MPa. (10 bar) – Quality: The quality of the compressed air should be in accordance with ISO 8573.1, class 2-3-2 [1)] • Electrical power installation (power to the test module): – Connection: 3 x 1.5 mm² cable – Voltage: 1-phase + neutral + earth (voltage variation from 187 to 264 V AC or from 93 to 132 V AC) – Frequency: 50/60 Hz • 12 V DC power supply (power to the CUC) • Earth net: – Max. transition resistance between earth wire and earth: ≤ 2.0 Ohm
Consumption data	• Compressed air [2)]: – Approx. 19.5 normal litres/operation • Electrical power: – < 100 w
Weight	• Approx. 600 kg
Static surface load	• < 0.25 N/mm² (2,5 kp/cm²)
Approvals	• The leak detector is designed for use in hazardous areas classified as Zone 1 according to IEC 79-10 and Class I, Division 1 according to NEC (USA), article 500
Other data	• The leak detector is designed for installation in locations in which the temperature may vary between -10°C and +40°C

[1)] The compressed air must not contain particles larger than 1 μm. The pressure dew point of the compressed air at a pressure of 0.7 MPa (7 bar) shall be at least 10°C lower than the lowest ambient temperature to which the compressed air system is exposed at the coldest place of the plant during the year (class 3 ~ -20°C). The maximum oil contents in the compressed air must not exceed 0.1 mg/m³.

[2)] The compressed air consumption can be used for dimensioning the compressor capacity (FAD = Free Air Delivery) based on local conditions. The specified compressed air consumption is based on the following standard reference conditions: An inlet pressure of 0.1 MPa (1 bar), a suction temperature of 20°C and a relative humidity of 60%.

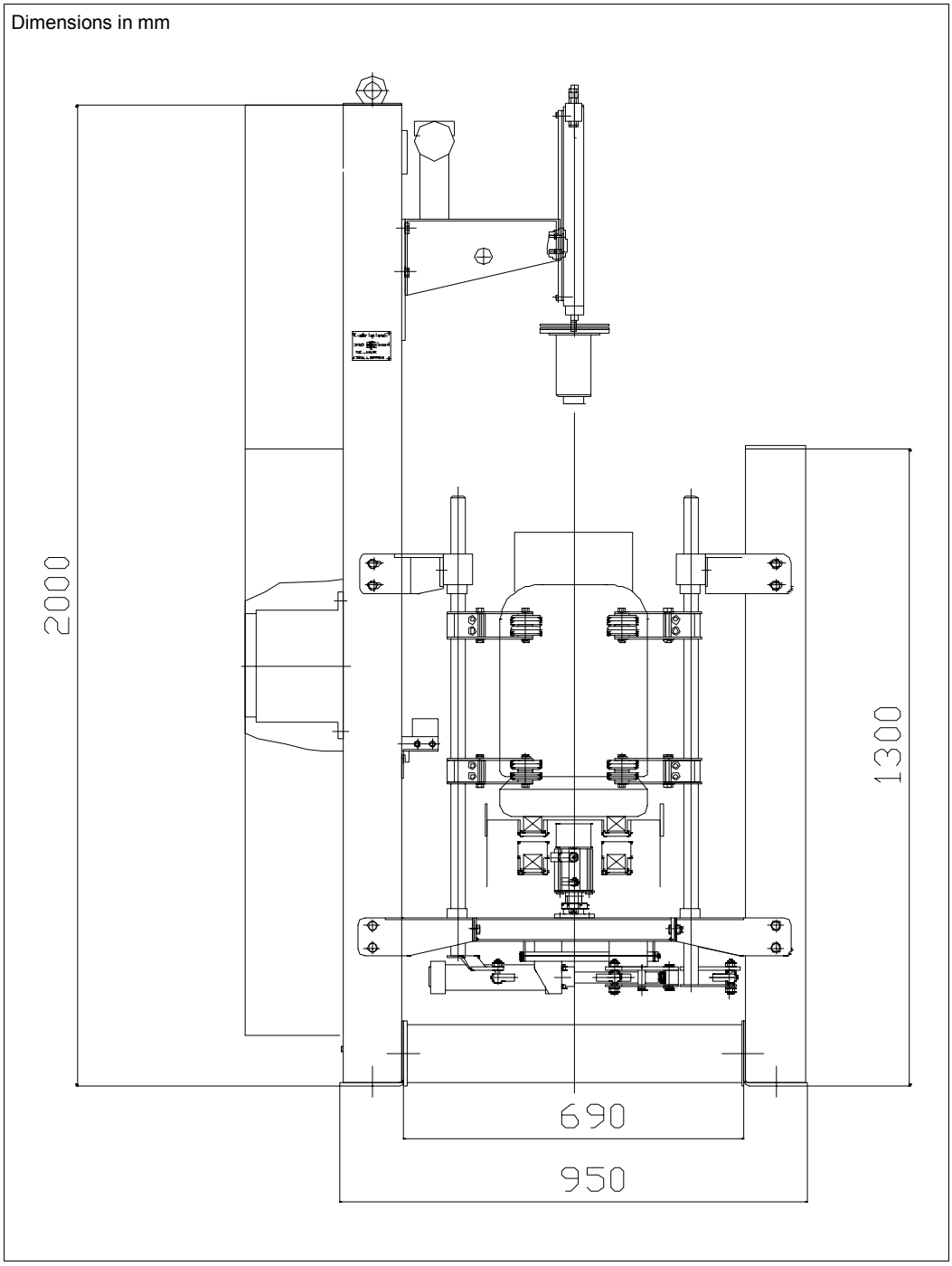
© Kosan Crisplant 2003

Dimensions in mm



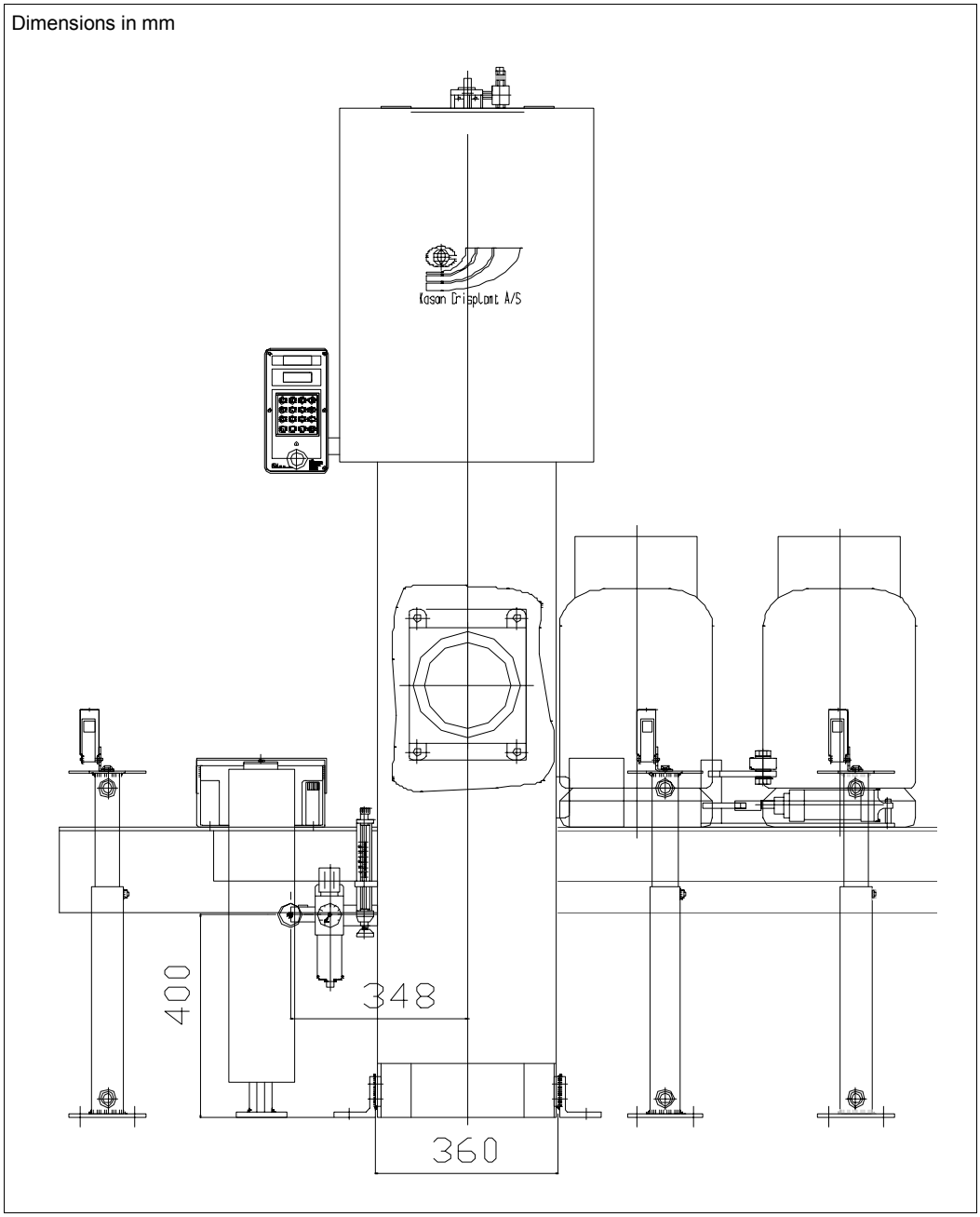
© Kosan Crisplant 2003

Dimensions in mm



© Kosan Crisplant 2003

Dimensions in mm



© Kosan Crisplant 2003